

Internship Project 2

Velocity and Distance Calculation of Electric Vehicle from Motor Torque using Simulink Model

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1 Abstract

Electric Vehicles (EVs) are rapidly emerging as the future of sustainable transportation, offering significant advantages over conventional internal combustion engine vehicles in terms of energy efficiency, environmental impact, and operational costs. A fundamental challenge in EV engineering is accurately predicting the velocity and distance that a vehicle can achieve given a specific motor torque output and a set of real-world resistive forces.

This project focuses on the development of a Simulink-based model to calculate the velocity and distance of an electric vehicle from its motor torque. The model accounts for the three primary resistive forces acting on a moving vehicle, namely the gradient force due to road inclination, the rolling resistance due to road surface friction, and the aerodynamic drag force due to air resistance. The net effective force acting on the vehicle is computed by subtracting the total resistive force from the motor-generated force, and this is used to determine acceleration, velocity, and distance through successive integration operations.

The Simulink block diagram environment of MATLAB is used to construct this model graphically, allowing a clear and intuitive representation of the physical system. The model is simulated with defined input parameters and the results show a final velocity of **93.71 KMPH** and a distance of **0.1302 KM**. This project demonstrates the effectiveness of Simulink as a tool for dynamic system modeling and simulation in electric vehicle engineering.

2 Introduction to SIMULINK Software

2.1 Overview of Simulink

Simulink is a graphical programming environment developed by MathWorks, integrated with MATLAB, and specifically designed for modeling, simulating, and analyzing dynamic systems. It provides a block diagram interface that allows engineers to represent complex physical and mathematical systems visually without writing traditional programming code.

Since its introduction, Simulink has become one of the most widely used simulation platforms in engineering disciplines including control systems, signal processing, communications, power electronics, and automotive engineering. Its ability to model both continuous and discrete time systems makes it particularly well-suited for real-world engineering simulations.

2.2 Key Features of Simulink

The following features of Simulink are relevant to this project:

- **Block Diagram Environment:** Simulink uses a drag-and-drop graphical interface where functional blocks are connected by signal lines to represent mathematical operations and system components. This makes it easy to visualize and understand the structure of a dynamic system.
- **Wide Block Library:** Simulink provides an extensive library of pre-built blocks for mathematical operations (sum, gain, product), signal sources (constants, step inputs), continuous systems (integrators, transfer functions), and display elements (scopes, displays). All of these are used in this project.
- **Integration with MATLAB:** Simulink operates within the MATLAB environment, allowing variables defined in the MATLAB workspace to be used directly in Simulink models. Results from Simulink simulations can be exported back to MATLAB for further analysis and plotting.
- **Real-Time Simulation:** Simulink supports real-time simulation and hardware-in-the-loop (HIL) testing, making it a powerful tool for validating control algorithms before deploying them on physical hardware.
- **Automatic Solver Selection:** Simulink automatically selects the appropriate numerical solver for the system being simulated, ensuring accurate results for both stiff and non-stiff differential equations.
- **Subsystem Creation:** Complex models can be organized into subsystems, making large models easier to manage, understand, and debug.

2.3 Why Simulink for EV Velocity and Distance Calculation

Simulink is particularly well-suited for this EV simulation project for the following reasons:

- The physical equations governing vehicle motion involve integration operations (acceleration to velocity, velocity to distance), which are natively represented by the Integrator block in Simulink.
- The graphical block diagram representation makes it easy to trace the flow of forces through the system, from torque input to final velocity and distance outputs.
- Multiple force components (gradient, rolling resistance, drag) can be computed in parallel branches and summed together, exactly mirroring the physical superposition of forces on the vehicle.
- Gain blocks allow easy scaling of signals by physical constants such as vehicle mass, wheel radius, and unit conversion factors.
- Display blocks provide immediate numerical readout of the simulation results without requiring additional post-processing.

3 Mathematical Equations for Motor Power and Torque System

3.1 Forces Acting on an Electric Vehicle

When an electric vehicle moves along a road, it experiences several resistive forces that oppose its motion. The motor must generate sufficient torque and force to overcome these resistances and accelerate the vehicle. The three primary resistive forces considered in this project are described below.

3.2 Gradient Force (Hill Climbing Force)

When a vehicle climbs an inclined road, it must overcome the component of its weight acting along the slope. This force is called the gradient force or hill climbing force, denoted by F_h .

$$F_h = W \sin \Phi = Mg \sin \Phi \quad (1)$$

Where:

- W = Weight of the vehicle = 2000 N
- M = Mass of the vehicle (kg)
- g = Acceleration due to gravity = 9.81 m/s^2
- Φ = Angle of inclination = 2.5 degrees

Calculation:

$$\begin{aligned} F_h &= 2000 \times \sin(2.5) \\ &= 2000 \times 0.04362 \\ &= \mathbf{87.23 \text{ N}} \end{aligned} \quad (2)$$

3.3 Rolling Resistance Force

Rolling resistance is the force resisting the motion of a vehicle rolling on a surface. It arises due to deformation of the tyre and the road surface at the contact patch.

$$F_r = C_r \times W \times \cos \Phi \quad (3)$$

Where:

- C_r = Rolling resistance coefficient = 0.004 (for asphalt roads)

- W = Weight of the vehicle = 2000 N
- Φ = Angle of inclination = 2.5 degrees

Calculation:

$$\begin{aligned}
 F_r &= 0.004 \times 2000 \times \cos(2.5) \\
 &= 0.004 \times 2000 \times 0.9990 \\
 &= \mathbf{7.99 \text{ N}}
 \end{aligned} \tag{4}$$

3.4 Aerodynamic Drag Force

The aerodynamic drag force opposes vehicle motion through air. It depends on the vehicle's speed, frontal area, drag coefficient, and air density.

$$F_d = 0.5 \times \rho \times C_d \times A \times V^2 \tag{5}$$

Where:

- ρ = Density of air = 1.2 kg/m³
- C_d = Drag coefficient = 0.5
- A = Frontal area of vehicle = 0.7 m²
- V = Vehicle velocity = 13.88 m/s (50 KMPH)

Calculation:

$$\begin{aligned}
 F_d &= 0.5 \times 1.2 \times 0.5 \times 0.7 \times 13.88 \times 13.88 \\
 &= 0.5 \times 1.2 \times 0.5 \times 0.7 \times 192.65 \\
 &= \mathbf{40.457 \text{ N}}
 \end{aligned} \tag{6}$$

3.5 Total Resistive Force

The total resistive force acting on the vehicle is the sum of all three individual forces:

$$F_{total} = F_h + F_r + F_d \tag{7}$$

Calculation:

$$\begin{aligned} F_{total} &= 87.23 + 7.99 + 40.457 \\ &= \mathbf{135.67 \text{ N}} \end{aligned} \quad (8)$$

3.6 Motor Torque and Force

The torque produced by the motor is related to the force at the wheel by the wheel radius:

$$T_{motor} = F_{motor} \times r_{wheel} \quad (9)$$

Therefore, the force generated by the motor is:

$$F_{motor} = \frac{T_{motor}}{r_{wheel}} \quad (10)$$

In this simulation, a motor torque of **200 Nm** is applied with a wheel radius of **0.3 m**, giving:

$$\begin{aligned} F_{motor} &= \frac{200}{0.3} \\ &= \mathbf{666.67 \text{ N}} \end{aligned} \quad (11)$$

3.7 Effective Force, Acceleration, Velocity and Distance

The effective force that actually accelerates the vehicle is:

$$F_{eff} = F_{motor} - F_{total} \quad (12)$$

$$\begin{aligned} F_{eff} &= 666.67 - 135.67 \\ &= \mathbf{531 \text{ N}} \end{aligned} \quad (13)$$

The acceleration of the vehicle is given by Newton's second law:

$$a = \frac{F_{eff}}{M} \quad (14)$$

Velocity is obtained by integrating acceleration with respect to time:

$$v = \int a \, dt \quad (15)$$

Distance is obtained by integrating velocity with respect to time:

$$d = \int v \, dt \quad (16)$$

These integration operations are implemented using the Integrator block in Simulink.

3.8 Summary of Design Parameters

Parameter	Value	Unit
Vehicle Weight	2000	N
Angle of Inclination	2.5	degrees
Rolling Resistance (C_r)	0.004	—
Drag Coefficient (C_d)	0.5	—
Frontal Area (A)	0.7	m ²
Air Density (ρ)	1.2	kg/m ³
Vehicle Velocity	13.88	m/s
Motor Torque	200	Nm
Wheel Radius	0.3	m
Gradient Force (F_h)	87.23	N
Rolling Force (F_r)	7.99	N
Drag Force (F_d)	40.457	N
Total Resistive Force	135.67	N
Motor Force	666.67	N
Effective Force	531	N

Table 1: Summary of design parameters and calculated values

4 Simulink Model for Velocity and Distance Calculation

4.1 Model Overview

The Simulink model was constructed to simulate the velocity and distance of an electric vehicle based on the motor torque input and the resistive forces acting on the vehicle. The model consists of four main branches that compute the gradient force, rolling resistance, aerodynamic drag, and motor force respectively. These are combined in a summing junction to obtain the effective force, which is then processed through gain and integrator blocks to produce the final velocity and distance outputs.

The simulation results show a steady-state velocity of **93.71 KMPH** and a distance of **0.1302 KM** for the given input parameters.

4.2 Screenshot 1 – Complete Simulink Model Overview

The following screenshot shows the complete Simulink block diagram model with all branches and connections visible. This gives a full picture of how the forces are calculated and combined to produce the velocity and distance outputs.

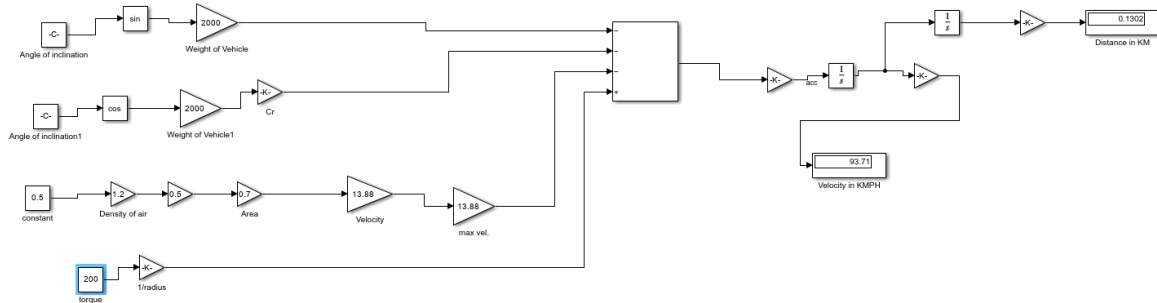


Figure 1: Complete Simulink Model – EV Velocity and Distance Calculation

4.3 Screenshot 2 – Gradient Force Calculation Block

This section of the model calculates the hill climbing force $F_h = W \sin \Phi$. The constant block feeds the angle of inclination (2.5 degrees) into a sine function block. The output is multiplied by the vehicle weight of 2000 N using a gain block to produce the gradient force of 87.23 N.

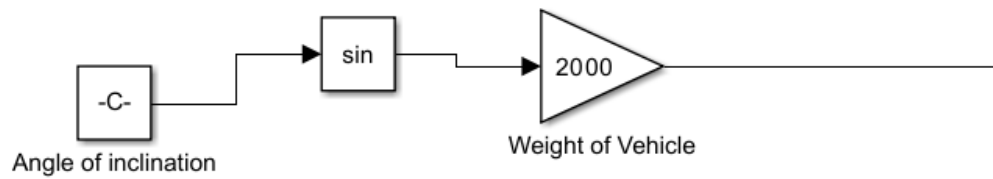


Figure 2: Gradient Force Block – $F_h = W \sin \Phi = 87.23 \text{ N}$

4.4 Screenshot 3 – Rolling Resistance Calculation Block

This branch calculates the rolling resistance force $F_r = C_r \times W \times \cos \Phi$. The angle of inclination is passed through a cosine function block, multiplied by the vehicle weight of 2000 N, and then scaled by the rolling resistance coefficient $C_r = 0.004$ using a gain block to give 7.99 N.

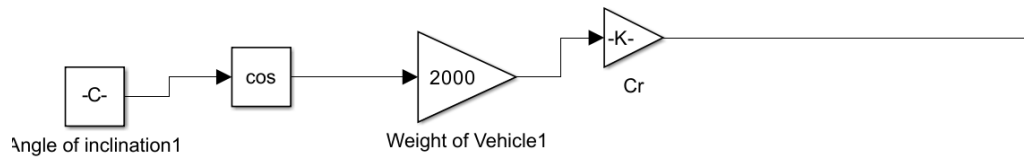


Figure 3: Rolling Resistance Block – $F_r = C_r \times W \times \cos \Phi = 7.99 \text{ N}$

4.5 Screenshot 4 – Aerodynamic Drag Force Calculation Block

This branch computes the aerodynamic drag force $F_d = 0.5 \times \rho \times C_d \times A \times V^2$. The constant 0.5 is multiplied through gain blocks representing air density (1.2), drag coefficient (0.5), frontal area (0.7), and velocity squared (13.88×13.88) to produce the drag force of 40.457 N.



Figure 4: Aerodynamic Drag Force Block – $F_d = 0.5\rho C_d A V^2 = 40.457 \text{ N}$

4.6 Screenshot 5 – Motor Torque Input Block

This branch represents the motor torque input. A constant block set to **200 Nm** feeds into a gain block representing the inverse of the wheel radius ($1/\text{radius} = 1/0.3$). This converts the motor torque into the equivalent force at the wheel, giving $F_{\text{motor}} = 666.67 \text{ N}$ which is fed into the summing junction as the positive driving force.



Figure 5: Motor Torque Input Block – Torque = 200 Nm, $F_{\text{motor}} = 666.67 \text{ N}$

4.7 Screenshot 6 – Summing Junction and Effective Force Block

The summing junction combines all four force inputs. The motor force (positive) is added and the three resistive forces (gradient, rolling, drag) are subtracted to compute the effective force acting on the vehicle. This effective force is then divided by the vehicle mass using a gain block to produce the acceleration signal.

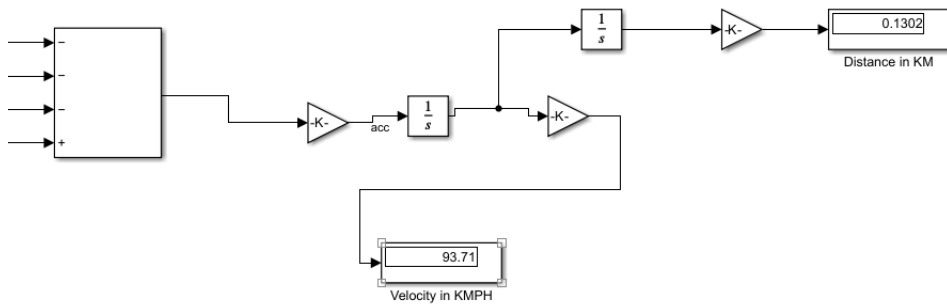


Figure 6: Summing Junction – Effective Force = $F_{motor} - F_{total} = 531 \text{ N}$

4.8 Screenshot 7 – Velocity Calculation Block

The acceleration signal from the previous stage is fed into an Integrator block ($1/s$) which performs time integration to obtain the vehicle velocity in m/s. A gain block then converts this to KMPH by multiplying by 3.6. The Display block shows the final steady-state velocity of **93.71 KMPH**.

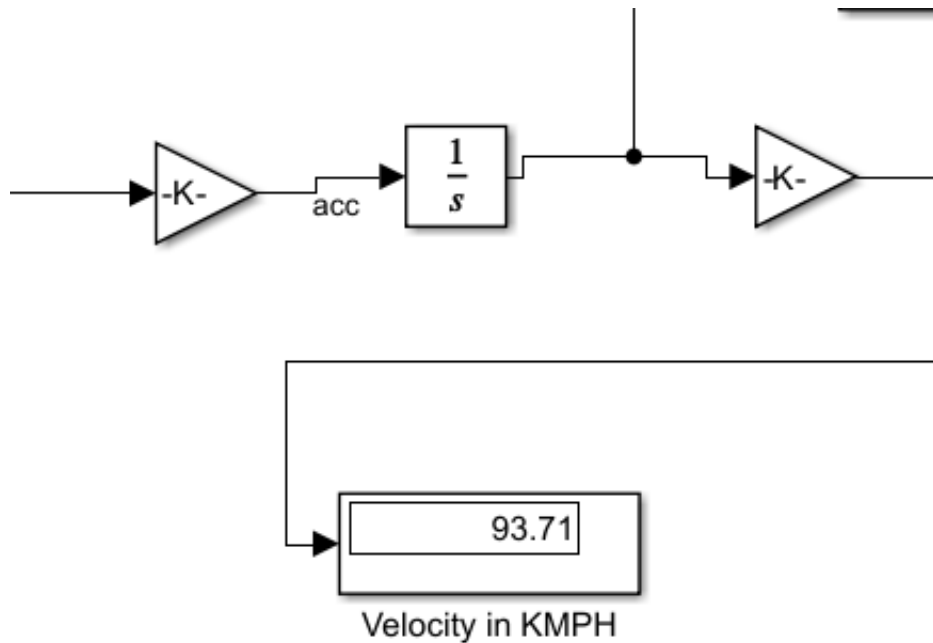


Figure 7: Velocity Calculation Block – Output Velocity = 93.71 KMPH

4.9 Screenshot 8 – Distance Calculation Block

The velocity signal (in m/s) is fed into a second Integrator block ($1/s$) which integrates velocity over time to compute the distance travelled by the vehicle. A gain block converts the output to kilometres by dividing by 1000. The Display block shows the final distance of **0.1302 KM**.

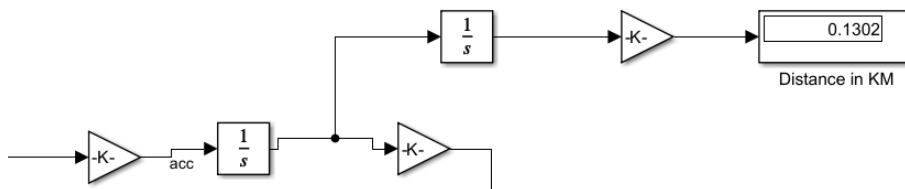


Figure 8: Distance Calculation Block – Output Distance = 0.1302 KM

5 Results and Conclusion

5.1 Simulation Results

The Simulink model was successfully simulated with the defined input parameters. The key results obtained from the simulation are summarized in the table below:

Output Parameter	Value	Unit
Gradient Force (F_h)	87.23	N
Rolling Resistance (F_r)	7.99	N
Aerodynamic Drag (F_d)	40.457	N
Total Resistive Force	135.67	N
Motor Force	666.67	N
Effective Force	531.00	N
Final Velocity	93.71	KMPH
Distance Travelled	0.1302	KM

Table 2: Summary of simulation results

5.2 Analysis of Results

The simulation results demonstrate that the Simulink model accurately captures the dynamic behavior of the electric vehicle under the specified operating conditions. The following observations are made:

- The gradient force of 87.23 N is the largest resistive force, accounting for approximately 64% of the total resistance. This highlights the significant impact of road inclination on vehicle performance, particularly for heavier vehicles.
- The aerodynamic drag force of 40.457 N is the second largest resistive force, contributing approximately 30% of total resistance. At higher speeds, this force will increase significantly since it is proportional to the square of velocity.
- The rolling resistance of 7.99 N is relatively small, contributing only about 6% of total resistance. This is consistent with the low rolling resistance coefficient of asphalt road surfaces.
- The motor force of 666.67 N is significantly larger than the total resistive force of 135.67

N, resulting in a large effective force of 531 N. This causes rapid acceleration of the vehicle, which explains the high final velocity of 93.71 KMPH achieved within the simulation time.

- The distance of 0.1302 KM travelled within the simulation period confirms that the vehicle accelerates quickly from rest and covers a reasonable distance, consistent with the high effective force and resulting acceleration.

5.3 Key Conclusions

The following key conclusions are drawn from this project:

- The Simulink model successfully demonstrates the complete force chain from motor torque input to final velocity and distance output, providing a clear and accurate representation of EV dynamics.
- The use of integrator blocks in Simulink to perform the conversion from acceleration to velocity and from velocity to distance perfectly mirrors the mathematical integration operations in the governing equations of motion, confirming the physical correctness of the model.
- Road inclination has the most significant impact on the resistive forces acting on an EV. Engineers must carefully account for the expected terrain when sizing the motor and battery for an electric vehicle.
- Aerodynamic drag increases with the square of velocity, meaning that at higher speeds the drag force will dominate the total resistance. This makes aerodynamic design increasingly important for high-speed EV applications.
- The Simulink block diagram approach provides a highly intuitive and transparent modeling framework where each physical component of the system corresponds directly to a specific block in the diagram, making the model easy to understand, modify, and extend.
- The model can be easily adapted to study the effect of changing any individual parameter such as motor torque, vehicle mass, road inclination, or drag coefficient, simply by adjusting the corresponding constant or gain block in the diagram.

5.4 Practical Implications for EV Engineering

The findings of this simulation have several important practical implications for electric vehicle design and engineering:

- **Motor Sizing:** The simulation shows that a motor torque of 200 Nm is more than sufficient to overcome the total resistive force of 135.67 N at the given conditions. In practice,

engineers must size the motor to handle the worst-case combination of gradient, rolling resistance, and drag at maximum speed.

- **Range Estimation:** The distance output of the Simulink model provides a direct method for estimating vehicle range under specific driving conditions. By varying the simulation time and torque input profile, the model can predict how far the vehicle will travel on a single battery charge.
- **Torque-Velocity Relationship:** The results confirm that higher motor torque directly translates to higher vehicle velocity and greater distance, but the relationship is modulated by the total resistive forces. Reducing aerodynamic drag or road gradient has a significant positive effect on vehicle performance.
- **Control System Design:** The model structure developed in this project forms the plant model that can be used to design and test velocity controllers, torque controllers, and regenerative braking systems in future work.

5.5 Future Scope

This project can be extended in several directions in future work:

- Incorporating a battery model to simulate the state of charge (SOC) and estimate vehicle range under different driving profiles.
- Adding a gear transmission model to simulate multi-speed drivetrains and their effect on torque-speed characteristics.
- Implementing a closed-loop velocity control system using a PID controller in Simulink to regulate vehicle speed to a desired set-point.
- Including regenerative braking in the model to recover kinetic energy during deceleration and improve overall system efficiency.
- Extending the model to simulate driving over a complete drive cycle such as the Worldwide Harmonised Light Vehicle Test Procedure (WLTP) to obtain realistic range and energy consumption estimates.
- Coupling the Simulink model with a thermal model to monitor motor and battery temperature during operation and evaluate the need for active cooling systems.

5.6 Final Remarks

This project successfully demonstrated the use of Simulink for modeling and simulating the dynamic behavior of an electric vehicle. The block diagram approach provided a clear, transparent, and physically meaningful representation of all the forces, accelerations, velocities, and

distances involved in EV motion. The simulation results of 93.71 KMPH velocity and 0.1302 KM distance confirm that the model behaves correctly and produces physically reasonable outputs for the given input parameters.

The skills developed in this project, including force analysis, equation of motion formulation, Simulink block diagram construction, and simulation result interpretation, form a strong foundation for more advanced work in electric vehicle dynamics, powertrain engineering, and control system design.